



Project Title - Seasonal Agriculture Performance Analysis

Student Name : **Samyak Jain**

College Name : **Delhi Technological University (DTU)**

AICTE STU ID : STU6a6cdf370fb751785519927



PROBLEM STATEMENT

- **Core Challenge:** Agricultural productivity, input efficacy, and farm profitability experience significant shifts due to seasonal environmental swings.
- **Research Focus:** Identify seasonal variations, explore trends across Kharif, Rabi, and Zaid seasons, and examine the impact of resource utilization on harvest outcomes.



Project Description

- **Analytical Scope:** Statistical analysis and exploratory data visualization of 4,000 farm records across diverse districts and farming setups.
- **Methodology:** Data cleaning with group median imputation, multivariate correlation analysis, resource efficiency assessment, and economic viability benchmarking across cropping cycles.

WHO ARE THE END USERS?

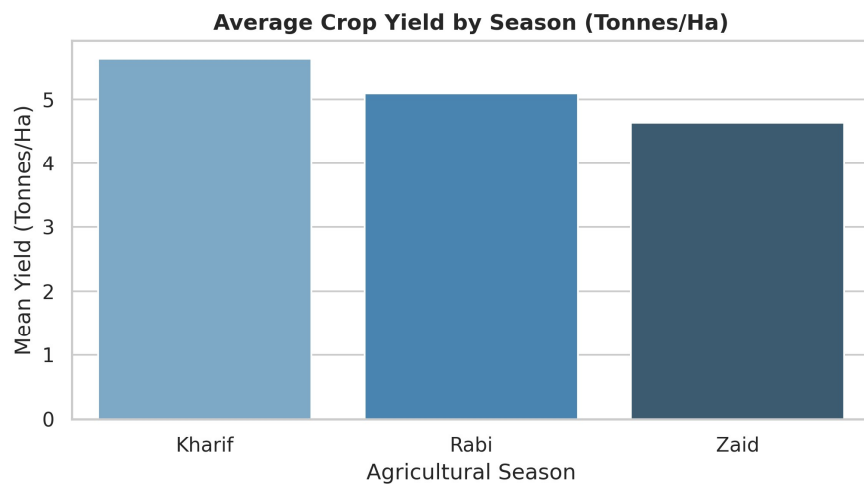
- 
- **Farmers & Farmer Producer Organizations (FPOs):** For choosing suitable crops and allocating fertilizer and irrigation budgets effectively by season.
 - **Agronomists & Policy Planners:** For setting regional irrigation priorities and formulating targeted fertilizer subsidy interventions.
 - **Agricultural Credit & Insurance Providers:** For designing seasonal underwriting models and identifying climate-related risks.
- 
- 

Technology Used

- **Language & Runtime:** Python 3 on Google Colab
- **Data Processing:** Pandas, NumPy
- **Data Visualization:** Matplotlib, Seaborn
- **Presentation & Documentation:** Microsoft PowerPoint



Results - Seasonal Crop Yield Trends



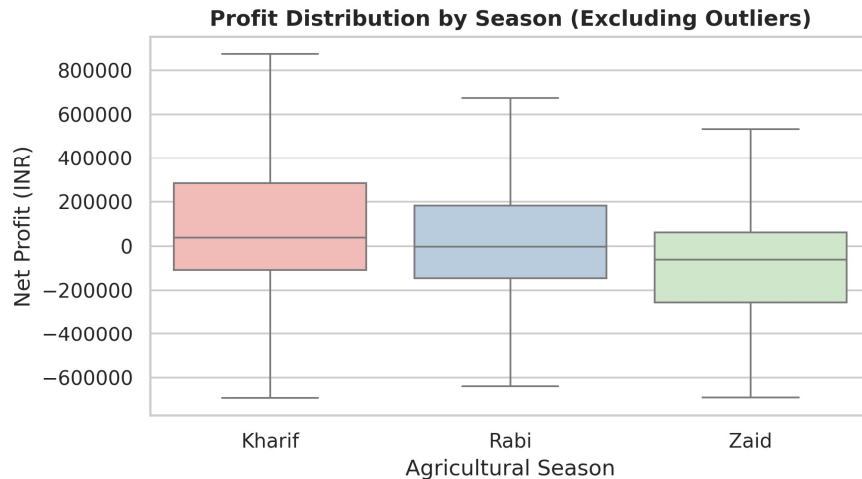
```
[3] ✓ Os
sns.set_theme(style="whitegrid")
plt.rcParams.update({'figure.autolayout': True})

plt.figure(figsize=(7, 4))
sns.barplot(data=df, x='Season', y='Yield Tonnes Ha', estimator=np.mean, errorbar=None, palette='Blues_d')
plt.title('Average Crop Yield by Season (Tonnes/Ha)', fontsize=12, fontweight='bold')
plt.xlabel('Agricultural Season')
plt.ylabel('Mean Yield (Tonnes/Ha)')
plt.savefig('chart1_seasonal_yield.png', dpi=300)
plt.show()
```

Key Observations:

- Kharif delivers the highest mean yield at **5.63 tonnes/ha**, supported by seasonal monsoon moisture.
- Rabi follows at **5.09 tonnes/ha**, while Zaid drops to **4.64 tonnes/ha** due to higher heat and reduced water retention.

Results - Economic Profitability by Season

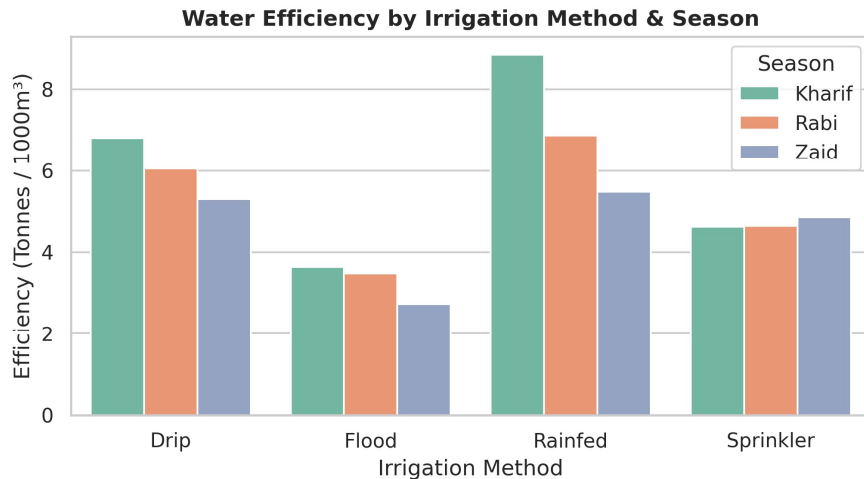


```
[4] 0s
plt.figure(figsize=(7, 4))
sns.boxplot(data=df, x='Season', y='Profit_INR', palette='Pastell', showfliers=False)
plt.title('Profit Distribution by Season (Excluding Outliers)', fontsize=12, fontweight='bold')
plt.xlabel('Agricultural Season')
plt.ylabel('Net Profit (INR)')
plt.savefig('chart2_profit_distribution.png', dpi=300)
plt.show()
```

Key Observations:

- Kharif achieves the strongest profitability, averaging ₹1,78,915 net profit per farm.
- Rabi records a moderate average profit of ₹87,689, whereas Zaid registers an average loss of -₹24,805 driven by high input and supplementary irrigation expenses.

Results - Water Efficiency & Irrigation Systems

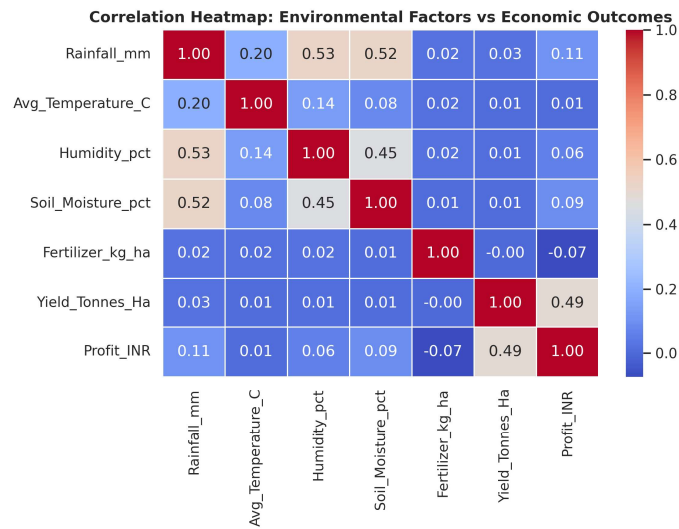


```
[5] plt.figure(figsize=(7, 4))
sns.barplot(data=df, x='Irrigation_Method', y='Water_Efficiency_t_per_1000m3', hue='Season', errorbar=None, palette='Set2')
plt.title('Water Efficiency by Irrigation Method & Season', fontsize=12, fontweight='bold')
plt.xlabel('Irrigation Method')
plt.ylabel('Efficiency (Tonnes / 1000m³)')
plt.savefig('chart3_irrigation_efficiency.png', dpi=300)
plt.show()
```

Key Observations:

- Drip irrigation yields the most stable output across seasons with high water-use efficiency.
- Traditional flood irrigation shows the lowest efficiency (3.44 t/1000m³), indicating high water loss through runoff and evaporation.

Results - Environmental & Economic Heatmap

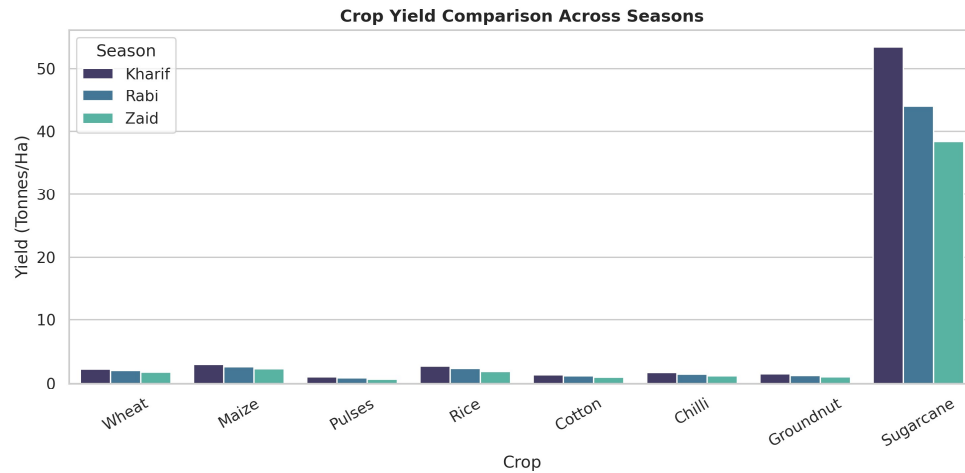


```
[6] ▶ plt.figure(figsize=(8, 6))
✓ 2s corr_cols = ['Rainfall_mm', 'Avg_Temperature_C', 'Humidity_pct', 'Soil_Moisture_pct', 'Fertilizer_kg_ha', 'Yield_Tonnes_Ha', 'Profit_INR']
sns.heatmap(df[corr_cols].corr(), annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Correlation Heatmap: Environmental Factors vs Economic Outcomes', fontsize=12, fontweight='bold')
plt.savefig('chart4_correlation_heatmap.png', dpi=300)
plt.show()
```

Key Observations:

- Soil moisture and fertilizer application rates correlate positively with final crop yield.
- Extreme temperatures and water stress correlate with reduced net margins, particularly outside the monsoon period.

Results - Crop Performance Comparison

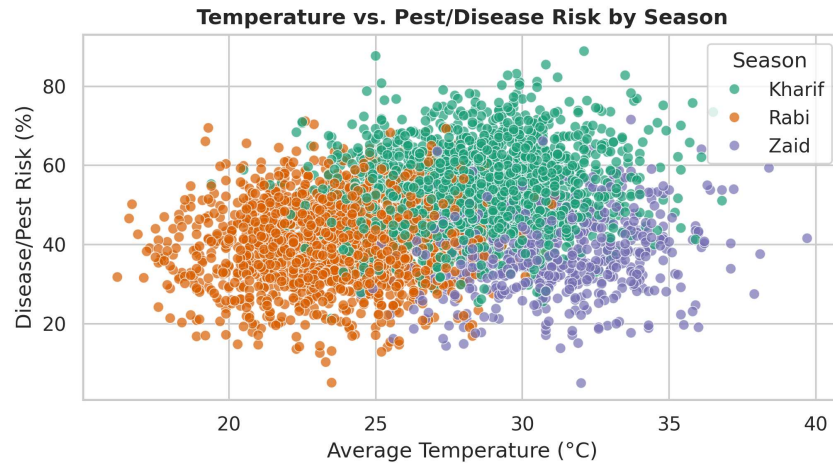


```
[7] In [7]: plt.figure(figsize=(10, 5))
sns.barplot(data=df, x='Crop', y='Yield_Tonnes_Ha', hue='Season', errorbar=None, palette='mako')
plt.title('Crop Yield Comparison Across Seasons', fontsize=12, fontweight='bold')
plt.ylabel('Yield (Tonnes/Ha)')
plt.xticks(rotation=30)
plt.savefig('chart5_crop_seasonal_yield.png', dpi=300)
plt.show()
```

Key Observations:

- Cash crops like Sugarcane show consistent volume differences across seasons (53.5 t/ha in Kharif vs. 38.4 t/ha in Zaid).
- Grains and pulses (Rice, Maize, Pulses) maintain peak productivity during Kharif and Rabi cycles.

Results - Temperature & Pest Risk Dynamics

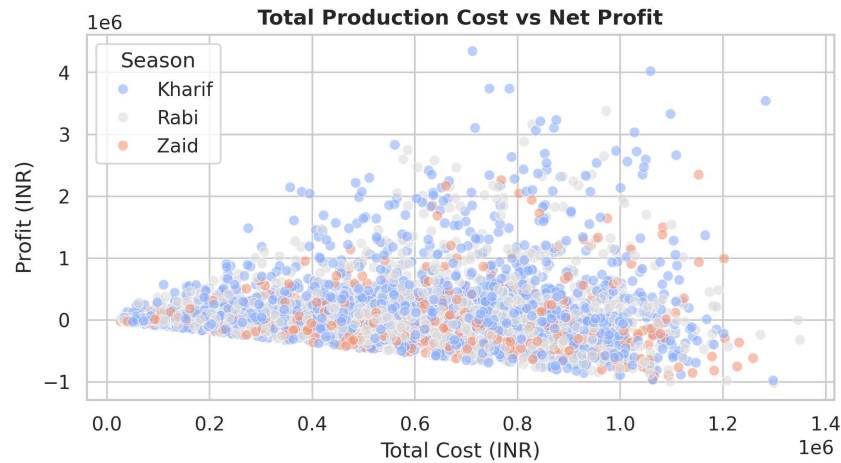


```
[8] ▶ plt.figure(figsize=(7, 4))
sns.scatterplot(data=df, x='Avg_Temperature_C', y='Disease_Pest_Risk_pct', hue='Season', palette='Dark2', alpha=0.7)
plt.title('Temperature vs. Pest/Disease Risk by Season', fontsize=12, fontweight='bold')
plt.xlabel('Average Temperature (°C)')
plt.ylabel('Disease/Pest Risk (%)')
plt.savefig('chart6_temp_pest_risk.png', dpi=300)
plt.show()
```

Key Observations:

- Disease and pest risk is highest during the high-humidity Kharif season (averaging **54.5%**).
- Rabi and Zaid exhibit lower average pest risks (**40.5%** and **38.2%**), but require targeted pesticide management during localized heat spikes.

Results - Production Cost vs. Net Profit

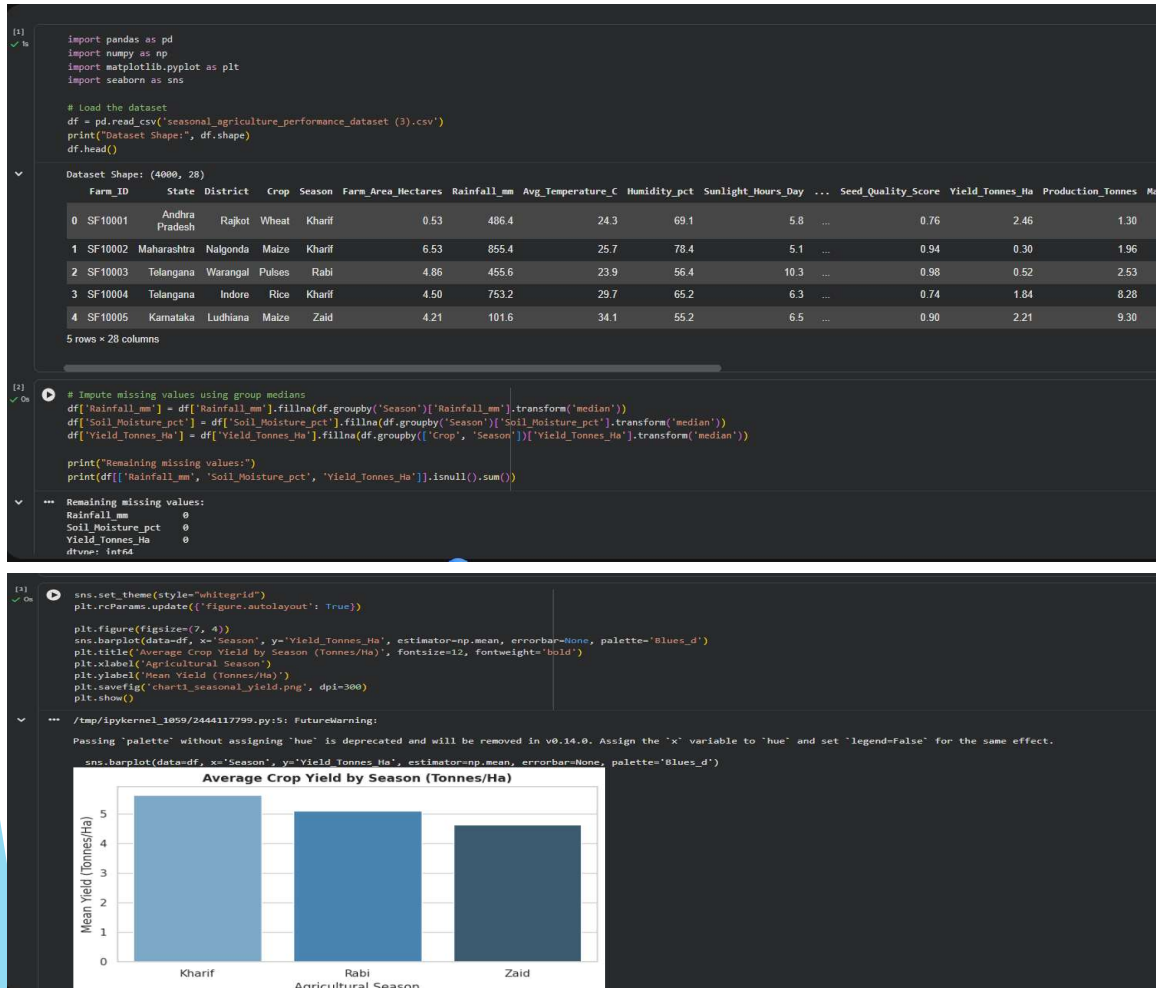


```
[9] plt.figure(figsize=(7, 4))
sns.scatterplot(data=df, x='Total_Cost_INR', y='Profit_INR', hue='Season', palette='coolwarm', alpha=0.6)
plt.title('Total Production Cost vs Net Profit', fontsize=12, fontweight='bold')
plt.xlabel('Total Cost (INR)')
plt.ylabel('Profit (INR)')
plt.savefig('chart7_cost_vs_profit.png', dpi=300)
plt.show()
```

Key Observations:

- Higher production expenditure does not ensure proportional net profit gains across all seasons.
- High-cost Zaid farming often yields lower margins, emphasizing the need for cost-efficient input budgeting.

Results - Colab Code Screenshots



Future scope

- **Predictive Yield Modeling:** Developing ML algorithms (e.g., Random Forest, XGBoost) to forecast crop output before planting.
- **IoT Weather Integration:** Combining real-time automated soil-moisture telemetry with targeted irrigation alerts.
- **Precision Nutrient Delivery:** Building algorithmic fertilizer blend recommendations based on soil pH and crop type.

GitHub Link



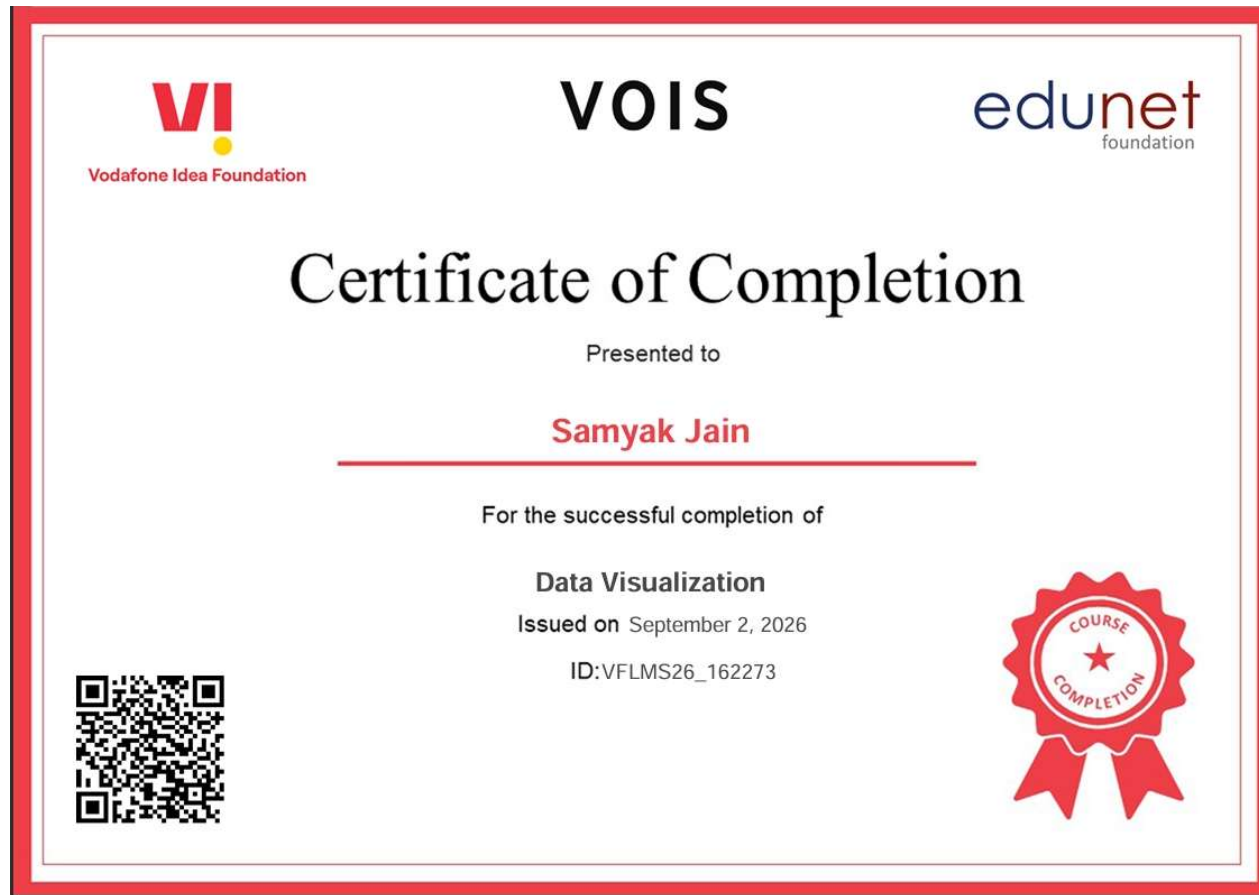
<https://github.com/samyakjain790-max/Seasonal-Agriculture-Performance-Analysis>

Contents Included : Colab notebook (.ipynb), cleaned dataset, Presentation PPT and generated visual plots.



VOIS Course completion certificate -

Course name- Data Visualization



THANK YOU



Samyak Jain

STU6a6cdf370fb751785519927